

Question Number	Answer	Additional guidance	Mark
5(a)(i)	• correct magnification calculated	ACCEPT range between x 14 000 - x 18 000 ACCEPT correct magnification in standard form	(1)

Question Number	Answer	Additional guidance	Mark
5(a)(ii)	An answer that makes reference to the following point: • too small ($0.5\ \mu\text{m}$) to be viewed (under a light microscope) / magnification (of x 14 000-16 000) can't be achieved by a light microscope / resolution would be too low} (1)	ecf applies ignore references to 3D image	(1)

Question Number	Answer	Additional guidance	Mark
5(b)	An explanation that makes reference to two of the following points: • plant based packaging is (more) sustainable / biodegradable / carbon neutral (than traditional oil-based packaging) (1) • (the antimicrobial property of the oregano oil) will {reduce the numbers of (<i>L. monocytogenes</i>) bacteria / prevent bacterial growth (on the food) / prevent microbial growth} (1) • (resulting in) fewer people getting {listeriosis / (microbial) disease} (1)	ACCEPT plant-based packaging is {renewable / available for future generations} ACCEPT kill bacteria ignore reduce microbes entering packaging ACCEPT so food will not go off as quick / extends shelf life	(2)
Question Number	Answer		Mark
5(c)(i)	The only correct answer is C <i>A is not correct because L. monocytogenes is least affected by the chemicals</i> <i>B is not correct because L. monocytogenes is least affected by the chemicals</i> <i>D is not correct because L. monocytogenes is least affected by the chemicals</i>		(1)

Question Number	Answer	Additional guidance	Mark
5(c)(ii)	• 6.14 / 6.1 / 6	<u>Example of calculation:</u> $11.1 + 0.63 = 11.73$ $6.0 - 0.41 = 5.59$ $11.73 - 5.59 = 6.14$	(1)

Question Number	Answer	Additional guidance	Mark
5(d)	A description that makes reference to the following points: • to test for side-effects (1) • to determine {safe / minimum /effective/ suitable} {dosage / concentration} (1) • to determine if it is more effective than {existing treatments / placebo} (1)	ignore to check for safety do not accept testing for side effects in context of {in-vitro/animal} ACCEPT to {determine optimum dosage / adjust dose} ACCEPT compare effectiveness with {existing treatments / placebo} ignore to see if it's effective	(3)